

Textual Description of the ERD

1. Introduction

The International Airport Database is designed to manage the main operations of an airport. The database stores information about airports, airlines, flights, passengers, bookings, boarding passes, baggage, baggage checks, and security checks.

The Entity-Relationship Diagram (ERD) shows the main entities of the system, their attributes, primary keys, foreign keys, and relationships between the entities.

2. Entities and Attributes

AIRPORT

The AIRPORT entity stores information about airports.

Attributes:

- airport_id – INT, Primary Key
- airport_name – VARCHAR
- country – VARCHAR
- state – VARCHAR
- city – VARCHAR
- created_at – DATETIME
- updated_at – DATETIME

The airport_id uniquely identifies each airport.

AIRLINE

The AIRLINE entity stores information about airline companies.

Attributes:

- airline_id – INT, Primary Key
- airline_code – VARCHAR, Unique
- name – VARCHAR
- country – VARCHAR
- created_at – DATETIME
- updated_at – DATETIME

The airline_id uniquely identifies each airline.

FLIGHT

The FLIGHT entity stores information about scheduled and actual flights.

Attributes:

- flight_id – INT, Primary Key
- departing_gate – VARCHAR
- arriving_gate – VARCHAR
- created_at – DATETIME
- updated_at – DATETIME
- airline_id – INT, Foreign Key
- departure_airport_id – INT, Foreign Key
- arrival_airport_id – INT, Foreign Key
- scheduled_departure – DATETIME
- scheduled_arrival – DATETIME
- actual_departure – DATETIME
- actual_arrival – DATETIME

The flight_id uniquely identifies each flight. The foreign keys connect a flight with an airline, a departure airport, and an arrival airport.

PASSENGER

The PASSENGER entity stores information about passengers.

Attributes:

- passenger_id – INT, Primary Key
- first_name – VARCHAR
- last_name – VARCHAR
- gender – VARCHAR
- date_of_birth – DATE
- country_of_citizenship – VARCHAR
- country_of_residence – VARCHAR
- passport_number – VARCHAR, Unique
- created_at – DATETIME
- updated_at – DATETIME

The passenger_id uniquely identifies each passenger.

BOOKING

The BOOKING entity stores information about flight bookings made by passengers.

Attributes:

- booking_id – INT, Primary Key
- flight_id – INT, Foreign Key
- status – VARCHAR
- booking_platform – VARCHAR
- created_at – DATETIME
- updated_at – DATETIME
- ticket_price – DECIMAL
- passenger_id – INT, Foreign Key

The booking_id uniquely identifies each booking. The flight_id and passenger_id connect the booking with a flight and a passenger.

BOOKING_CHANGE

The BOOKING_CHANGE entity stores information about changes made to a booking.

Attributes:

- booking_change_id – INT, Primary Key
- booking_id – INT, Foreign Key
- change_details – TEXT
- created_at – DATETIME

The booking_id connects each change with the corresponding booking.

BOARDING_PASS

The BOARDING_PASS entity stores information about boarding passes issued to passengers.

Attributes:

- boarding_pass_id – INT, Primary Key
- seat – VARCHAR
- boarding_time – DATETIME
- created_at – DATETIME

- updated_at – DATETIME
- booking_id – INT, Foreign Key

The booking_id connects the boarding pass with a booking.

BAGGAGE

The BAGGAGE entity stores information about passenger baggage.

Attributes:

- baggage_id – INT, Primary Key
- weight_kg – DECIMAL
- created_at – DATETIME
- updated_at – DATETIME
- booking_id – INT, Foreign Key

The booking_id connects baggage with the corresponding booking.

BAGGAGE_CHECK

The BAGGAGE_CHECK entity stores information about baggage checking at the airport.

Attributes:

- baggage_checking_id – INT, Primary Key
- check_results – VARCHAR
- created_at – DATETIME
- updated_at – DATETIME
- booking_id – INT, Foreign Key
- passenger_id – INT, Foreign Key

The booking_id and passenger_id connect the baggage check with a booking and a passenger.

SECURITY_CHECK

The SECURITY_CHECK entity stores information about passenger security checks.

Attributes:

- security_check_id – INT, Primary Key
- check_results – VARCHAR
- created_at – DATETIME
- updated_at – DATETIME

- `passenger_id` – INT, Foreign Key

The `passenger_id` connects the security check with the passenger.

3. Relationships

AIRLINE – FLIGHT

The relationship between AIRLINE and FLIGHT is One-to-Many (1:M). One airline can have many flights, while each flight belongs to one airline.

AIRPORT – FLIGHT

The relationship between AIRPORT and FLIGHT is One-to-Many (1:M). An airport can be the departure airport for many flights. An airport can also be the arrival airport for many flights.

The FLIGHT entity has two foreign keys for these relationships: `departure_airport_id` and `arrival_airport_id`.

FLIGHT – BOOKING

The relationship between FLIGHT and BOOKING is One-to-Many (1:M). One flight can have many bookings, while each booking is associated with one flight.

PASSENGER – BOOKING

The relationship between PASSENGER and BOOKING is One-to-Many (1:M). One passenger can make many bookings, while each booking belongs to one passenger.

BOOKING – BOOKING_CHANGE

The relationship between BOOKING and BOOKING_CHANGE is One-to-Many (1:M). One booking can have multiple changes recorded, while each booking change belongs to one booking.

BOOKING – BOARDING_PASS

The relationship between BOOKING and BOARDING_PASS is One-to-One (1:1) in this simplified database model. A booking can have one boarding pass, and each boarding pass belongs to one booking.

BOOKING – BAGGAGE

The relationship between BOOKING and BAGGAGE is One-to-Many (1:M). One booking can have multiple baggage records, while each baggage record belongs to one booking.

PASSENGER – BAGGAGE_CHECK

The relationship between PASSENGER and BAGGAGE_CHECK is One-to-Many (1:M). One passenger can have multiple baggage check records, while each baggage check is associated with one passenger.

PASSENGER – SECURITY_CHECK

The relationship between PASSENGER and SECURITY_CHECK is One-to-Many (1:M). One passenger can have multiple security check records, while each security check belongs to one passenger.

4. Keys and Constraints

Primary keys are used to uniquely identify records in each entity. Foreign keys are used to connect related entities and maintain data integrity.

The AIRLINE entity uses airline_id as its primary key, while airline_code is unique. The PASSENGER entity uses passenger_id as its primary key, while passport_number is unique.

The FLIGHT entity contains foreign keys to AIRLINE and AIRPORT. The BOOKING entity contains foreign keys to FLIGHT and PASSENGER. Other entities also use foreign keys to connect their records with the appropriate parent entities.

5. ERD Legend

- PK – Primary Key
- FK – Foreign Key
- UK – Unique Key
- 1:M – One-to-Many relationship
- 1:1 – One-to-One relationship
- DATETIME – Date and time value
- VARCHAR – Text value
- INT – Integer number
- DATE – Date value
- DECIMAL – Decimal number
- TEXT – Long text value

6. Conclusion

The ERD represents the main structure of the International Airport Database. It shows how airports, airlines, flights, passengers, bookings, boarding passes, baggage, and different types of checks are connected. Primary keys identify individual records, while foreign keys establish relationships between the entities. The database structure allows airport information and passenger-related operations to be stored in an organized way.

